

ALGORITHMIC NON-COMMUTATIVE CLASS FIELD THEORY, XIII: UNIVERSAL LOCALIZATION OF THE DEGENERACY ALGEBRA, THE K_1 -CHARACTERISTIC CLASS, AND THE EISENSTEIN BOUNDARY

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ABSTRACT. Problem 5.1 of [V] and its residue [VII, Rem. 3.1] asked for a single class in K_1 of a localized degeneracy algebra whose evaluations reproduce the snake factorization of [VII]: the arc term $q^{n_k(q-1)}$ and the Eisenstein index q at every level of the Iwahori tower. Ore localization is unavailable because the degeneracy algebra is non-commutative and non-normal ([V, Prop. 4.1]). This paper constructs the class with Cohn’s universal localization, and finds that the construction reorganizes the whole picture. The degeneracy order $\mathcal{H}$ is the $\mathbb{Z}_p$ -algebra generated by the four degeneracy maps $\tau_*, \tau^*, \eta_*, \eta^*$ acting on the augmentation-zero lattices $\mathcal{M}_k^0 = \ker(\deg) \subset \mathbb{Z}_p^{n_k}$; the tail Laplacian factors inside it as $\Delta_k = \tau_*(\tau^* - \eta^*)$, tail norm times degeneracy coboundary. On $\mathcal{M}_k^0$ the Laplacian is injective with finite cokernel, so $\Delta = \bigoplus_k \Delta_k$ is Σ -inverting for $\Sigma = \{\Delta\}$; the universal localization $\mathcal{H} \rightarrow \mathcal{H}_\Sigma$ exists, the low-degree localization sequence $K_1(\mathcal{H}) \rightarrow K_1(\mathcal{H}_\Sigma) \xrightarrow{\partial} K_0(\mathcal{H}, \Sigma) \rightarrow K_0(\mathcal{H})$ holds, and $[\Theta_{\mathcal{H}}] := [\Delta] \in K_1(\mathcal{H}_\Sigma)$ is the characteristic class. Its content is in two theorems. The level evaluations $\text{ev}_k: K_1(\mathcal{H}_\Sigma) \rightarrow \mathbb{Q}_p^\times$ give $\text{ev}_k[\Theta_{\mathcal{H}}] = \pm \text{pdet}(\Delta_k) = \pm n_k \kappa(Y_k)$, and on K_4 these have 2-adic order exactly $n_k - 3$ at $k = 1, 2, 3$: the K_1 class carries the pure arc growth μp^k and nothing else. The boundary class $\partial[\Theta_{\mathcal{H}}]$ at level k is the cokernel of Δ_k on $\mathcal{M}_k^0$, which sits in a non-split extension $0 \rightarrow \mathbb{Z}/n_k \rightarrow \text{coker } \Delta_k^0 \rightarrow \text{Jac}(Y_k) \rightarrow 0$: the Euler term $-k = -\text{ord}_p n_k + \text{const}$ of [III] is the Eisenstein subclass $[\mathbb{Z}/n_k]$ of the boundary. Axioms (A1) and (A2) of [V] are thus separated inside a single K -theoretic object, and (A3) holds on abelian towers, where $\text{pdet}(\Delta_k) = n\kappa(Y_0) \prod_{\chi \neq 1} \det D(\chi)$ is the characteristic element of [IV] evaluated at all nontrivial characters (exact on the $(1, 1, 1)$ tower). What the specification called the “non-commutative main conjecture” $\partial[\Theta] = [X]$ is the definition of ∂ ; the genuine main conjecture — an analytic p -adic L -function of the tower, defined independently, equal to $[\Theta_{\mathcal{H}}]$ — is stated and left open.

STATUS OF THE RESULTS

Every numbered statement is proved. The external inputs are the existence and universal property of Cohn’s universal localization [2] and the low-degree localization exact sequence for universal localizations [4, Ch. 4]; the higher sequence of Neeman–Ranicki [3] requires stable flatness, which we do not verify and do not use. Everything else is finite linear algebra over $\mathbb{Z}_p$, machine-verified exactly on the K_4 Iwahori tower at $k \leq 3$ and on the $(1, 1, 1)$ abelian tower (Table 1). Section 5 says what the main conjecture is and is not.

1. THE DEGENERACY ORDER AND THE AUGMENTATION-ZERO LATTICES

Setting as in [V], [VII]: Y_k the Iwahori path tower over a $(q+1)$ -regular base, $q = p$, $n_k = n(q+1)q^{k-1}$, A_k the level Hecke operator, $\Delta_k = qI - A_k$, $\text{Tor}_k = \text{Jac}(Y_k) = \text{Tor coker } \Delta_k$; $\tau_*, \eta_* \in$

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Paper XIII of the series; Papers I–XII are cited as [I]–[XII]. The verification script `cohn_localization.py` in the ancillary bundle reproduces every computational claim; its output is archived as `cohn_localization.output.txt`.

$M_{n_k \times n_{k+1}}(\mathbb{Z})$ the tail and head pushforwards, τ^*, η^* their transposes, with [V, Prop. 1.2]

$$(1.1) \quad \tau_* \eta^* = A_k, \quad \eta^* \tau_* = A_{k+1}, \quad \tau_* \tau^* = \eta_* \eta^* = qI_{n_k},$$

and $\deg: \mathbb{Z}^{n_k} \rightarrow \mathbb{Z}$ the coordinate sum.

Lemma 1.1 (Degree compatibility and the Laplacian factorization). $\deg \circ \tau_* = \deg \circ \eta_* = \deg$ and $\deg \circ \tau^* = \deg \circ \eta^* = q \deg$; hence all four degeneracy maps preserve the augmentation-zero lattices $\mathcal{M}_k^0 := \ker(\deg) \subset \mathbb{Z}_p^{n_k}$. Moreover

$$\Delta_k = \tau_* (\tau^* - \eta^*) = \tau_* \partial_k^*,$$

the tail norm composed with the degeneracy coboundary ∂_k^* of [V, Thm. 2.1]; in particular Δ_k lies in the algebra generated by the degeneracy maps and maps $\mathbb{Z}_p^{n_k}$ into $\mathcal{M}_k^0$.

Proof. Pushforwards sum over fibres and pullbacks have q elements per fibre. By (1.1), $\tau_* \tau^* - \tau_* \eta^* = qI - A_k$. Column sums of Δ_k vanish (in-degree q). Verified exactly at $k = 1, 2, 3$ (Table 1, lines (I),(B)). $\square$

Definition 1.2 (Degeneracy order). For $K \geq 1$ let $\mathcal{M}_{\leq K}^0 := \bigoplus_{k \leq K+1} \mathcal{M}_k^0$, and let $\mathcal{H}_K \subset \text{End}_{\mathbb{Z}_p}(\mathcal{M}_{\leq K}^0)$ be the $\mathbb{Z}_p$ -subalgebra generated by the degeneracy maps between consecutive levels $\leq K+1$, restricted to the augmentation-zero lattices by Lemma 1.1. It is an order in a semisimple $\mathbb{Q}_p$ -algebra (a $\mathbb{Z}_p$ -subalgebra of a matrix algebra, finitely generated as a module), non-commutative and non-normal since $A_k A_k^t \neq A_k^t A_k$ inside it [V, Prop. 4.1]. Put

$$\Delta_{(K)} := \bigoplus_{k \leq K} \Delta_k^0 \oplus q \cdot \text{id}_{\mathcal{M}_{K+1}^0} \in \mathcal{H}_K, \quad \Delta_k^0 := \Delta_k|_{\mathcal{M}_k^0},$$

where the last summand $q \text{id} = \tau_* \tau^*$ keeps the element inside $\mathcal{H}_K$. We write $\mathcal{H}$ for $\mathcal{H}_K$ with K fixed and large, and $\Sigma := \{\Delta_{(K)}\}$.

Proposition 1.3 (Σ -invertible). Δ_k^0 is injective with finite cokernel and invertible over $\mathbb{Q}_p$, with

$$\det \Delta_k^0 = \pm \text{pdet}(\Delta_k) = \pm n_k \kappa(Y_k),$$

the product of the nonzero eigenvalues of Δ_k . Hence $\Delta_{(K)} \otimes \mathbb{Q}_p$ is invertible while $\Delta_{(K)}$ is not invertible in $\mathcal{H}$, its determinant not being a $\mathbb{Z}_p$ -unit: localization is needed and is non-trivial.

Proof. $\ker \Delta_k = \mathbb{Z}_p \mathbf{1}$ and $\deg \mathbf{1} = n_k \neq 0$, so Δ_k is injective on $\ker(\deg)$; Δ_k maps $\mathbb{Z}_p^{n_k}$ into $\ker(\deg)$, so Δ_k^0 is an endomorphism of the rank- $(n_k - 1)$ lattice $\mathcal{M}_k^0$ with finite cokernel. Its determinant is the product of the eigenvalues of Δ_k on the complement of the kernel, i.e. the pseudo-determinant, which equals $n_k \kappa(Y_k)$ by [III, eq. (2.2)]. On K_4 : $4608 = 12 \cdot 384$, $18874368 = 24 \cdot 786432$, $316659348799488 = 48 \cdot 6597069766656$, with 2-adic orders 9, 21, 45. $\square$

2. UNIVERSAL LOCALIZATION AND THE CHARACTERISTIC CLASS

Definition 2.1 (Universal localization and the relative group). By Cohn [2] there is a ring homomorphism $\lambda: \mathcal{H} \rightarrow \mathcal{H}_\Sigma$, universal among ring homomorphisms under which $\Delta_{(K)}$ becomes invertible. Let $K_0(\mathcal{H}, \Sigma)$ be the relative group of triples (P, Q, f) , P, Q finitely generated projective $\mathcal{H}$ -modules and $f: P \rightarrow Q$ a map with $\lambda(f)$ invertible, modulo the usual relations [4, Ch. 4]. Then

$$(2.1) \quad K_1(\mathcal{H}) \longrightarrow K_1(\mathcal{H}_\Sigma) \xrightarrow{\partial} K_0(\mathcal{H}, \Sigma) \longrightarrow K_0(\mathcal{H})$$

is exact [4, Ch. 4], and ∂ sends the class of $\lambda(f)$, for $f: P \rightarrow Q$ with $\lambda(f)$ invertible, to $[(P, Q, f)]$. Forgetting to modules, $[(P, Q, f)] \mapsto [\text{coker } f] - [\text{ker } f]$ when these are finite.

Definition 2.2 (Characteristic class). $[\Theta_{\mathcal{H}}] := [\lambda(\Delta_{(K)})] \in K_1(\mathcal{H}_\Sigma)$. It is well defined by Proposition 1.3, and $\mathcal{H}_\Sigma \neq 0$ because λ extends to the representation $\mathcal{H} \rightarrow \text{End}_{\mathbb{Q}_p}(\mathcal{M}_{\leq K}^0 \otimes \mathbb{Q}_p)$ in which $\Delta_{(K)}$ is invertible.

Theorem 2.3 (The tautological identity, and what it says). $\partial[\Theta_{\mathcal{H}}] = [(\mathcal{M}_{\leq K}^0, \mathcal{M}_{\leq K}^0, \Delta_{(K)})] \in K_0(\mathcal{H}, \Sigma)$, whose image under the forgetful map is

$$\sum_{k \leq K} [\text{coker } \Delta_k^0] + [\mathcal{M}_{K+1}^0 / q \mathcal{M}_{K+1}^0].$$

This is the definition of ∂ ; it is the statement the specification called the “non-commutative main conjecture”, and it has no content beyond Definition 2.1. The content is in Theorems 3.1 and 3.2.

3. EVALUATION AND THE EISENSTEIN BOUNDARY

For each $k \leq K$ the projection $\mathcal{H} \rightarrow \text{End}_{\mathbb{Z}_p}(\mathcal{M}_k^0)$ followed by $\otimes \mathbb{Q}_p$ is a ring homomorphism $\mathcal{H} \rightarrow M_{n_k-1}(\mathbb{Q}_p)$ under which $\Delta_{(K)}$ becomes invertible; by universality it factors through $\mathcal{H}_{\Sigma}$, and composing with the determinant gives the *level evaluation*

$$\text{ev}_k : K_1(\mathcal{H}_{\Sigma}) \longrightarrow K_1(M_{n_k-1}(\mathbb{Q}_p)) = \mathbb{Q}_p^{\times}.$$

Theorem 3.1 (Level evaluations: pure arc growth). $\text{ev}_k[\Theta_{\mathcal{H}}] = \det \Delta_k^0 = \pm n_k \kappa(Y_k)$, and for all $k \geq 1$

$$\frac{\text{ev}_{k+1}[\Theta_{\mathcal{H}}]}{\text{ev}_k[\Theta_{\mathcal{H}}]} = \pm q^{n_k(q-1)}, \quad \text{ord}_p \text{ev}_k[\Theta_{\mathcal{H}}] = n_k + c = \frac{n(q+1)}{q} p^k + c,$$

with a constant c depending only on the base. The arc term $\text{Det}_{\text{arc}} = q^{n_k(q-1)}$ of [VII, Cor. 2.2] is the ratio of consecutive evaluations of the single class $[\Theta_{\mathcal{H}}]$; the Euler term does not appear. On K_4 : $c = -3$ and $\text{ord}_2 \text{ev}_k = 9, 21, 45$ at $k = 1, 2, 3$.

Proof. The first equality is Proposition 1.3. The ratio: $\text{pdet}(\Delta_{k+1}) / \text{pdet}(\Delta_k) = q^{n_{k+1}-n_k} = q^{n_k(q-1)}$ because the nonzero spectrum of A_{k+1} is that of A_k plus $n_{k+1}-n_k$ zeros [III, eq. (2.1)], each contributing the eigenvalue q of Δ_{k+1} . Equivalently this is the residue constancy $n_k \kappa_k q^{-n_k} = \text{const}$ of [IX, Thm. 4.1(iii)]. Summing, $\text{ord}_p \text{ev}_k = n_k + c$ with $n_k = n(q+1)q^{k-1}$. $\square$

Theorem 3.2 (The Eisenstein boundary). For every $k \geq 1$ there is an exact sequence of finite $\mathbb{Z}_p$ -modules

$$0 \longrightarrow \mathbb{Z}/n_k \longrightarrow \text{coker } \Delta_k^0 \longrightarrow \text{Jac}(Y_k) \longrightarrow 0,$$

so that in the Grothendieck group of finite $\mathbb{Z}_p$ -modules the level- k component of $\partial[\Theta_{\mathcal{H}}]$ is

$$[\text{coker } \Delta_k^0] = [\text{Jac}(Y_k)] + [\mathbb{Z}/n_k], \quad \text{ord}_p |\text{Jac}(Y_k)| = \text{ord}_p \text{ev}_k[\Theta_{\mathcal{H}}] - \text{ord}_p n_k = \frac{n(q+1)}{q} p^k - k + \nu_0.$$

The Euler term $-k$ of [III, Thm. 2.2] is the Eisenstein subclass $[\mathbb{Z}/n_k]$ of the boundary, of p -adic order $\text{ord}_p n_k = k + \text{ord}_p(n(q+1)) - 1$. On K_4 the extension is non-split at $k = 1, 2, 3$: $\text{coker } \Delta_k^0 = \mathbb{Z}/8 \oplus (\mathbb{Z}/24)^2$, $(\mathbb{Z}/2)^9 \oplus \mathbb{Z}/16 \oplus (\mathbb{Z}/48)^2$, $(\mathbb{Z}/2)^{12} \oplus (\mathbb{Z}/4)^9 \oplus \mathbb{Z}/32 \oplus (\mathbb{Z}/96)^2$, against $\text{Jac}(Y_k) \oplus \mathbb{Z}/n_k$, whose invariant factors differ (at $k = 1$: $\mathbb{Z}/2 \oplus \mathbb{Z}/4 \oplus (\mathbb{Z}/24)^2$).

Proof. $\Delta_k(\mathbb{Z}_p^{n_k}) \subseteq \ker(\deg)$, so $\text{Jac}(Y_k) = \text{Tor coker } \Delta_k = \ker(\deg) / \Delta_k(\mathbb{Z}_p^{n_k})$ by [VII, Lem. 1.1] and [IV, Lem. 1.3]. The lattice $\Delta_k(\mathbb{Z}_p^{n_k})$ contains $\Delta_k(\ker \deg)$ with quotient $\Delta_k(\mathbb{Z}_p^{n_k}) / \Delta_k(\ker \deg) \cong \mathbb{Z}_p^{n_k} / (\ker \deg + \mathbb{Z}_p \mathbf{1}) \cong \mathbb{Z} / \deg(\mathbf{1}) = \mathbb{Z}/n_k$, since Δ_k is injective modulo $\mathbb{Z}_p \mathbf{1}$. The exact sequence follows, and orders multiply: $|\text{coker } \Delta_k^0| = n_k |\text{Jac}(Y_k)|$, consistent with Proposition 1.3. The p -adic law is Theorem 3.1 minus $\text{ord}_p n_k$; on K_4 , $n_k - 3 - (k+1) = 6 \cdot 2^k - k - 4$, the law of [III, Thm. 2.2]. Non-splitting: compare invariant factors (Table 1, line (B)). $\square$

Remark 3.3 (Axioms (A1), (A2), (A3) in K -theory). [V, §5] asked for a characteristic formalism producing an arc-density term μ (A1), an Euler term $-\chi k$ intrinsic to the base (A2), and the commutative characteristic ideal on abelian sub-towers (A3); [VII] delivered all three in evaluated form and left the lift to a single K_1 class as the residue. Theorems 3.1–3.2 are that lift: the class $[\Theta_{\mathcal{H}}]$ evaluates to the pure arc law (A1); its boundary contains the Eisenstein subclass $[\mathbb{Z}/n_k]$ whose

order is the vertex count, and $-\text{ord}_p n_k$ is the Euler term (A2) — the fourth reading of $-\chi k$ after [III], [VII] and [IX], and the first in which it is a class rather than a number; (A3) is Theorem 3.4. The three axioms are properties of one object, not three formulas.

Theorem 3.4 (Abelian reduction). *For an abelian $\mathbb{Z}/\ell^k$ voltage tower of [IV] with pencil $D(t)$ and $\Delta_k = D(S_{\ell^k})$ [IV, Lem. 1.1], the same construction (the group ring $\mathbb{Z}_\ell[G_k]$ playing the role of $\mathcal{H}$) gives*

$$\text{ev}_k[\Theta] = \text{pdet}(\Delta_k) = n_k \kappa(Y_k) = n \kappa(Y_0) \prod_{\chi \neq 1} \det D(\chi) = n \kappa(Y_0) \prod_{\zeta^{\ell^k}=1, \zeta \neq 1} \zeta^{-d} g(\zeta - 1),$$

the characteristic element $g = \det D(1+T)$ of [IV, Thm. 2.1] evaluated at all nontrivial characters, times the Eisenstein constant $n \kappa(Y_0)$; the Eisenstein subclass is $\mathbb{Z}/n_k = \mathbb{Z}/n \ell^k$ and $\text{ord}_\ell n_k = k + \text{ord}_\ell n$ is the “ $-k$ ” of [II, Rem. 5.4]. Exact on the $(1, 1, 1)$ tower, $\ell = 2$, $k \leq 3$: 3072, 2408448, 362577395712 (Table 1, line (A3)).

Proof. The block-character decomposition [IV, Lem. 1.1] splits the nonzero spectrum of Δ_k into the spectra of $D(\chi)$, $\chi \neq 1$, and the nonzero spectrum of $D(1) = \Delta_0$, whose product is $n \kappa(Y_0)$ [III, eq. (2.2)]; $\det D(\zeta) = \zeta^{-d} g(\zeta - 1)$ as in [IX, Lem. 2.1]. $\square$

4. THE VERIFICATION BATTERY

check	levels	result
(I) (1.1); $A_k A_k^t \neq A_k^t A_k$; degree compatibility of $\tau_*, \eta_*, \tau^*, \eta^*$	$k = 1, 2, 3$	exact
(S) Δ_k^0 injective, invertible over $\mathbb{Q}$, det not a $\mathbb{Z}_2$ -unit (2-adic orders 9, 21, 45)	$k = 1, 2, 3$	exact
(E) $\det \Delta_k^0 = n_k \kappa_k = \text{pdet}$; ratios $2^{12}, 2^{24}$; $\text{ord}_2 = n_k - 3$	$k = 1, 2, 3$	exact (pdet numeric)
(B) $ \text{coker } \Delta_k^0 / \text{Jac}(Y_k) = n_k$; structures; extension non-split; $\Delta_k = \tau_*(\tau^* - \eta^*)$	$k = 1, 2, 3$	exact
(A3) $(1, 1, 1)$, $\ell = 2$: $\text{pdet}(\Delta_k) = n \kappa_0 \prod_{\chi \neq 1} \det D(\chi)$	$k = 1, 2, 3$	exact

TABLE 1. Verification (`cohn_localization.py`) on the K_4 Iwahori tower and the $(1, 1, 1)$ abelian tower.

5. SCOPE, RESIDUE, CORRECTIONS

Remark 5.1 (What a main conjecture would be). In the theory of [1] a main conjecture asserts that an *analytic* object — a p -adic L -function characterized by interpolation of special values, constructed independently of the module — equals the algebraic characteristic element, an element of K_1 of the localized Iwasawa algebra whose boundary is the class of the Selmer module. The algebraic half is what this paper constructs: $[\Theta_{\mathcal{H}}] \in K_1(\mathcal{H}_{\Sigma})$ with $\partial[\Theta_{\mathcal{H}}]$ the class of the tower’s cokernels, the analogue of “ $\partial \xi = [X]$ ”. The analytic half does not exist yet: there is no independently defined element of $K_1(\mathcal{H}_{\Sigma})$ interpolating the L -values of the tower — the candidates are the pseudo-determinants $n_k \kappa_k = \prod_j (q+1 - \lambda_j)$, products of local L -values at the Eisenstein point as in [I, Thm. 3.5], but a construction of a single element from them, rather than from Δ , is exactly what is missing. The non-commutative main conjecture for Hecke towers is therefore: *construct $\mathcal{L}_{\mathcal{H}} \in K_1(\mathcal{H}_{\Sigma})$ from the L -values alone and prove $\mathcal{L}_{\mathcal{H}} = [\Theta_{\mathcal{H}}]$ modulo the image of $K_1(\mathcal{H})$* . Its evaluated form is [VII, Thm. 2.1]; its K_1 form is open.

Remark 5.2 (Solved, open, impossible). *Solved*: the residue of [V, Prob. 5.1] and [VII, Rem. 3.1] in the form asked — a single class in K_1 of a localized degeneracy algebra whose level evaluations and boundary reproduce the snake factorization, with (A1) and (A2) separated as evaluation and Eisenstein subclass and (A3) as the abelian reduction. *Open*: (a) the analytic side (Remark 5.1);

(b) stable flatness of $\mathcal{H} \rightarrow \mathcal{H}_\Sigma$ and hence the higher localization sequence [3]; (c) the structure of $\mathcal{H}_\Sigma$ itself — whether it is an order in a semisimple algebra or has torsion, and the image of $K_1(\mathcal{H})$ in $K_1(\mathcal{H}_\Sigma)$, which measures how far $[\Theta_\mathcal{H}]$ is from unique; (d) the rank-two version over the degeneracy algebra of [XII], where no abelian sector exists. *Impossible or empty as stated*: “ $\partial[\Theta] = [X]$ ” as a theorem with content (Theorem 2.3).

Remark 5.3 (Corrections to the task specification). (1) The identity $\partial([\Theta_\mathcal{H}]) = [X_\infty]$ is the definition of the boundary map in the relative sequence, not a theorem; the paper proves what does have content (Theorems 3.1–3.4). (2) The class must be formed on the augmentation-zero lattices: on $\mathbb{Z}_p^{n_k}$ the Laplacian has the Eisenstein kernel and is not Σ -inverting; the specification’s “ $\Delta_\infty \in M_n(\widehat{\mathcal{H}}_\infty)$ ” has no $n \times n$ presentation because n_k grows (contrast [IV], where $\mathbb{Z}_\ell[G_k]^n$ is free of constant rank). (3) The Euler term is not produced by “the connecting map” inside K_1 ; it is the Eisenstein subclass $[\mathbb{Z}/n_k]$ of the boundary, and it is invisible to the K_1 -evaluations, which see the arc term only. (4) “ $\Sigma =$ the set of morphisms inducing surjections with finite cokernel on X_∞ ” is replaced by the single element $\Delta_{(K)}$; universal localization at one element already gives the class. (5) The higher K -theory sequence needs stable flatness [3]; only the low-degree sequence [4] is used, and only its first map matters for the construction.

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